

IEEE Transactions on Emerging Topics in Computational Intelligence

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AUTHOR GUIDELINES

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Article Type	Full Paper		
Title	Deep Learning Models for the Identification of Biomarker, Drug-Target Interaction, and Pharmacokinetics in Pancreatic Ductal Adenocarcinoma		
Manuscript Files	Name	Type of File	Size
	IEEE Transactions DL14-06-2026.docx	Anonymized Main Document - MS Word	947.3 KB
	IEEE Transactions DL14-06-2026.pdf	Anonymized Main Document - PDF	1.1 MB
	Deep Learning Models for the Identification of	Title Page	15 KB

Name	Type of File	Size
Biomarker.docx		
Supplementary_DL_PDA_C.docx	Supplementary Material for Review	760.5 KB
Cover Letter IEEE.docx	Cover letter / Comments	129.5 KB

Abstract

Pancreatic ductal adenocarcinoma (PDAC) develops in the ductal cells of the pancreas, which are responsible for carrying digestive enzymes. It is considered highly aggressive due to late diagnosis, poor survival rate, early metastasis, mostly unresectable, and responds poorly to current treatments. In this study, we attempted to identify the targets of PDAC, drug-target interaction (DTI), and pharmacokinetics of the identified drugs using deep learning models. Initially, we considered three microarray gene expression datasets publicly available in the NCBI GEO and pre-processed using a standard pipeline. We identified the differentially expressed genes between normal and PDAC using deep learning models such as long short-term memory (LSTM), recurrent neural network (RNN), and one-dimensional convolutional neural network (1D-CNN). Further, the drug library was prepared for identified targets using the DepMap database. The embeddings of targets were generated using ProtVec, ProtBert, evolutionary scale modeling (ESM), and drugs were made using Mol2Vec, graph isomorphism network (GIN), and Unimol. We performed the DTI analysis and pharmacokinetics of the identified drugs using the same deep learning models. Our study identified 22 biomarker genes with a balanced accuracy of 98.89% using RNN. We found 106 drugs based on DTI analysis with a high R^2 of 0.86 using Mol2Vec+GIN+Unimol+ProtVec+ProtBert+ESM and RNN. Our models were able to predict the pharmacokinetics properties such as absorption, distribution, metabolism, excretion, and toxicity with a mean absolute error of 1.62 (GIN+Mol2Vec+RNN), Spearman correlation (SC) of 0.56 (Mol2Vec+LSTM), area under the curve of 62.5% (Mol2Vec+Unimol+1D-CNN), SC of 0.32 (Unimol+LSTM), and 0.36 (GIN+Mol2Vec+1D-CNN), respectively. Our study demonstrates the potential of deep learning models as an effective and scalable solution for prognostic marker identification, disease-specific drug-target interaction

prediction, and pharmacokinetics prediction, with promising implications for drug repurposing in PDAC.

Authors

Title	Name	Email	Country/Location
Miss	Pragya Pragya ¹	pragya.rs.bme22@iitbhu.ac.in	India
Mr.	Aditya Acharya ¹	aditya.acharya.bme23@iitbhu.ac.in	India
Mr.	Konatala Vibhuvan Krishna ¹	kvibhuvan.krishna.bme22@iitbhu.ac.in	India
Mr.	Abhijeet Chahar ¹	abhijeet.chahar.bme21@iitbhu.ac.in	India
Dr.	Bhuvaneshwari Balasubramaniam ²	bhuvana.chy@iitbhu.ac.in	India
Dr.	Jac Fredo Agastinose Ronickom ¹ Corresponding Author Submitting Author	jack.bme@iitbhu.ac.in	India

[0000-0001-5759-6632](#)

Affiliations

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1. Computational Neuroscience and Biology Lab, School of Biomedical Engineering, Indian Institute of Technology (BHU) Varanasi, 221005, India
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Funders	Is this research supported by funding? • I-DAPT Hub Foundation, Indian Institute of Technology (BHU) Varanasi Grant numbers • No. I-DAPT/IIT (BHU)/2025-26/ Project Sanction/51
Additional Information	Keywords Pancreatic ductal adenocarcinoma; Biomarker; Drug target interaction; Pharmacokinetics; Deep learning Has this manuscript been submitted previously to this journal? Yes, this is a resubmission TETCI-2026-1514 Informed Consent No, there were no human subjects Animal Subjects There were no animal subjects Opposed Reviewers <i>Not provided</i>

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History	Submitted On	14 June 2026 by Jac Fredo Agastinose Ronickom
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	Submission Started	14 June 2026 by Jac Fredo Agastinose Ronickom